

PROSPERO Registration — Completed Submission Draft

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Every PROSPERO field is drafted below so you can copy each answer into the matching field of the online form. Text in **red** marks the few items to confirm with Prof. Basso and the student before submitting. Register before formal screening begins.

1. Review title

Combination and adjunctive treatment regimens for *Pneumocystis jirovecii* pneumonia (PCP) in HIV-negative / immunocompromised adults: a systematic review and Bayesian meta-analysis.

2. Original language title

Trattamento di combinazione e adiuvante della polmonite da *Pneumocystis jirovecii* (PCP) nei pazienti adulti HIV-negativi / immunocompromessi: revisione sistematica e meta-analisi bayesiana.

3. Anticipated or actual start date

1 July 2026.

↳ **Confirm:** Use the date the review work actually began; searches were finalised 23 July 2026.

4. Anticipated completion date

31 December 2026.

↳ **Confirm:** Align with the intended thesis defence session (autumn/winter 2026).

5. Stage of review at time of this submission

Review stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Searches and search-piloting are complete; the 1,560 de-duplicated records have not yet been screened. Update these flags to the position on the actual submission date.

6. Named contact

Russell Edward Lewis.

↳ **Confirm:** Confirm the registered named contact (guarantor) for the review.

7. Named contact email

russelledward.lewis@unipd.it

8. Named contact address

Department of Molecular Medicine (Dipartimento di Medicina Molecolare), University of Padua, Via A. Gabelli 63, 35121 Padua, Italy.

↳ **Confirm:** Verify the exact departmental postal address.

9. Named contact phone (optional)

[Optional — add a contact telephone number if desired.]

10. Organisational affiliation of the review

University of Padua (Università degli Studi di Padova), Department of Molecular Medicine.

11. Review team members and their organisational affiliations

- Paulina Kullaku — Medical student (thesis candidate) and first reviewer, University of Padua.
- Prof. Monica Basso — Associate Professor of Infectious Diseases, Department of Molecular Medicine, University of Padua (supervisor / relatrice).
- Russell Edward Lewis — Infectious Diseases, University of Padua (supervisor / relatore; methodological lead and second reviewer).
- Nicola Bonadiman — University of Padua. **confirm role/affiliation (co-supervisor / resident).**

12. Funding sources / sponsors

This review is an academic thesis project and received no specific external funding. It is supported by the institutional resources of the University of Padua.

↳ **Confirm:** *Confirm no external funding.*

13. Conflicts of interest

None declared.

↳ **Confirm:** *Confirm none for all team members.*

14. Collaborators (optional)

None.

15. Review question

In HIV-negative / immunocompromised adults with *Pneumocystis jirovecii* pneumonia (PCP), what are the comparative effectiveness and safety of combination and adjunctive treatment regimens, compared with standard monotherapy (typically trimethoprim–sulfamethoxazole, TMP-SMX) or alternative single-agent regimens?

16. Searches

MEDLINE/PubMed, Embase (Ovid) and Scopus were searched from 2016 onward using a high-sensitivity, three-block strategy: (1) pathogen — *Pneumocystis* / *P. jirovecii* / *P. carinii* / PCP / PJP; AND (2) drugs/regimen — trimethoprim–sulfamethoxazole, co-trimoxazole, clindamycin, primaquine, atovaquone, pentamidine, dapsone, caspofungin/echinocandins, corticosteroids, adjunctive/combotherapy; AND (3) population — HIV-negative / non-HIV / immunocompromised / transplant / malignancy / haematologic / solid organ. No study-design filter was applied (sensitivity maximised; filtered at screening).

Records retrieved (search run 23 July 2026): PubMed 492, Embase (Ovid) 971, Scopus 904 — 2,367 in total; 1,560 records remained after de-duplication in Zotero. Screening will be performed in Covidence. A supervisor verification search will be run to confirm totals before screening.

↳ **Confirm:** *State whether Cochrane CENTRAL was searched as a separate source (some records are tagged Cochrane CENTRAL via the Ovid export). Add the final line-by-line strategy and search dates as a supplementary file.*

17. URL to search strategy / uploaded file

The full line-by-line search strategy for each database will be uploaded as a supplementary file with this registration (no external URL).

18. Condition or domain being studied

Pneumocystis jirovecii pneumonia (PCP/PJP) — an opportunistic fungal infection carrying substantial morbidity and mortality (reported 20–50% in non-HIV populations) in immunocompromised, HIV-negative hosts.

19. Participants / population

Inclusion: Adults (≥18 years) with proven or probable PCP who are HIV-negative / non-HIV immunocompromised — e.g., solid-organ transplant, haematopoietic stem cell transplant, haematological malignancy, solid tumour on chemotherapy, or autoimmune/rheumatologic disease on immunosuppression or corticosteroids.

Exclusion: HIV-positive-only populations; patients with colonisation but without a diagnosis of PCP; populations receiving prophylaxis rather than treatment.

↳ **Confirm:** *Age scope — restrict to adults, or also include paediatric studies? Paediatric records are present in the search results.*

20. Intervention(s) / exposure(s)

Combination or adjunctive treatment regimens for acute PCP — for example, TMP-SMX combined with a second agent (clindamycin–primaquine, atovaquone, pentamidine, dapsone, caspofungin/echinocandins) and/or adjunctive corticosteroids.

21. Comparator(s) / control

Standard-of-care monotherapy (typically TMP-SMX), an alternative single-agent regimen, or — where relevant — standard-dose versus reduced-dose comparisons.

↳ **Confirm:** *Confirm whether reduced- vs standard-dose comparisons are in scope.*

22. Types of study to be included

Include: Randomised controlled trials and comparative observational studies (prospective or retrospective cohort, case–control) reporting clinical outcomes by treatment group.

Exclude: Case reports and small case series, single-arm studies without a comparator, narrative and systematic reviews / meta-analyses (reference lists screened for completeness), editorials, letters, conference abstracts without extractable data, and in-vitro or animal studies.

23. Context

Adults with PCP managed in any healthcare setting (predominantly hospital / inpatient, including intensive care), in any country, from 2016 onward.

24. Main outcome(s)

All-cause mortality — measured at 30 days and/or in-hospital. Effect measures: risk ratio or odds ratio with 95% credible intervals (and absolute risk difference).

↳ **Confirm:** *Confirm the exact primary time point(s) and preferred effect measure.*

25. Additional outcome(s)

- Treatment failure / change of therapy.
- Respiratory failure or need for non-invasive / invasive mechanical ventilation.
- ICU admission.
- Length of hospital stay.
- Adverse events (Grade ≥3) and treatment discontinuation due to toxicity.

↳ **Confirm:** *Confirm the additional-outcome list, time points and effect measures.*

26. Data extraction (selection and coding)

Records will be imported into Covidence. Title/abstract and full-text screening will be performed independently by two reviewers, with disagreements resolved by discussion or by a third reviewer. Data will be extracted into a piloted

standardised form capturing study design, setting, population and immunosuppression type, intervention and comparator regimens, outcomes, follow-up, and risk-of-bias items; extraction will be checked by a second reviewer.

27. Risk of bias (quality) assessment

Risk of bias will be assessed independently by two reviewers using Cochrane RoB 2 for randomised trials and ROBINS-I for non-randomised comparative studies; disagreements resolved by discussion or a third reviewer.

↳ **Confirm:** *Confirm the tools.*

28. Strategy for data synthesis

A Bayesian random-effects meta-analysis will be performed; pooled effect estimates will be reported with 95% credible intervals. Where multiple regimens can be compared, a Bayesian network meta-analysis will be considered. Between-study heterogeneity will be assessed and explored; small-study effects will be examined where the number of studies permits. Analyses will use R (e.g., brms / Stan or JAGS).

↳ **Confirm:** *Confirm software, prior specification, and whether NMA is planned.*

29. Analysis of subgroups or subsets

Planned subgroup analyses (data permitting): by underlying immunosuppression (solid-organ transplant vs haematological malignancy vs autoimmune/other), by disease severity (e.g., respiratory support at baseline), and by study design (RCT vs observational).

↳ **Confirm:** *Confirm the subgroup list.*

30. Type and method of review

Systematic review with meta-analysis (Bayesian); network meta-analysis considered if data permit. Health area: Infection / Infectious diseases.

31. Language

No language restriction; non-English articles will be translated where feasible.

↳ **Confirm:** *Confirm — or restrict to English-language publications only.*

32. Country

Italy.

33. Other registration details

None. This review is not registered on any other register.

34. Reference / URL for published protocol

None; no separate protocol has been published. (This PROSPERO record serves as the registered protocol.)

35. Dissemination plans

Findings will be presented in the candidate's degree thesis (University of Padua) and are intended for submission to a peer-reviewed journal and, where appropriate, presentation at a scientific conference.

36. Keywords

Pneumocystis jirovecii pneumonia; PCP; PJP; HIV-negative; immunocompromised; combination therapy; adjunctive corticosteroids; trimethoprim–sulfamethoxazole; systematic review; Bayesian meta-analysis.

37. Existing reviews of the same topic by the same authors

The team is aware of related prior work on reduced-dose TMP-SMX for PCP identified during scoping. The present review addresses a distinct question (combination and adjunctive regimens) and is not a duplicate.

↳ **Confirm:** State explicitly whether any team member has authored a prior/related review, to avoid duplication concerns.

38. Current review status

Ongoing — searches completed; screening not yet started.

39. Any additional information

This is a degree thesis project at the University of Padua. A shared Zotero library holds the 1,560 de-duplicated records; study selection and data extraction will be managed in Covidence.

40. Details of final report / publication(s)

To be completed on review completion (thesis reference and any resulting publication with web link).

Complete draft for the PCP combination-therapy systematic review. Confirm the red items, then enter each answer into the corresponding PROSPERO field and submit before screening begins.